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ORIGINAL RESEARCH

Predicting Treatment Outcomes in Patients with Low Back Pain Using Gene Signature-Based Machine Learning Models

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ABSTRACT

***Introduction*:** Low back pain (LBP) is a significant global health burden, with variable treatment outcomes and an unclear underlying molecular mechanism. Effective prediction of treatment responses remains a challenge. In this study, we aimed to develop gene signature-based machine learning models using transcriptomic data from peripheral immune cells to predict treatment outcomes in patients with LBP.

***Methods*:** The transcriptomic data of patients with LBP from peripheral immune cells were retrieved from the GEO database. Patients with LBP were recruited, and treatment outcomes were assessed after 3 months. Patients were classified into two groups: those with resolved pain and those with persistent pain. Differentially expressed genes (DEGs) between the two groups were identified through bioinformatic analysis. Key genes were selected using five machine learning models, including Lasso, Elastic Net, Random Forest, SVM, and GBM. These key genes were then used to train 45 machine learning models by combining nine different algorithms: Logistic Regression, K-Nearest Neighbors, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting Machine, Multilayer Perceptron, Naive Bayes, and Linear Discriminant Analysis. Five-fold cross-validation was employed to ensure robust model evaluation and minimize overfitting. In each fold, the dataset was split into training and validation sets, with model performance assessed using multiple metrics including accuracy, precision, recall, and F1 score. The final model performance was reported as the mean and standard deviation across all five folds, providing a more reliable estimate of the models' ability to predict LBP treatment outcomes using gene expression data from peripheral immune cells.

***Results*:** A total of 61 DEGs were identified between patients with resolved and persistent pain. From these genes, 45 machine learning models were constructed using different

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combinations of feature selection methods and classification algorithms. The Elastic Net with Logistic Regression achieved the highest accuracy of 88.7% ± 8.0% (mean ± standard deviation), followed closely by Elastic Net with Linear Discriminant Analysis (88.7% ± 7.5%) and Lasso with Multilayer Perceptron (87.7% ± 6.7%). Overall, 15 models demonstrated robust performance with accuracy > 80%, suggesting the reliability of our machine learning approach in predicting LBP treatment outcomes. The SHapley Additive exPlanations (SHAP) method was used to visualize the contribution of core genes to model performance, highlighting their roles in predicting treatment outcomes.

Conclusion: The study demonstrates the potential of using transcriptomic data from peripheral immune cells and machine learning models to predict treatment outcomes in patients with LBP. The identification of key genes and the high accuracy of certain models provide a basis for future personalized treatment strategies in LBP management. Visualizing gene importance with SHAP adds interpretability to the predictive models, enhancing their clinical relevance.

Keywords: Low back pain; Treatment prediction; Gene signature; Machine learning; Transcriptomics; Peripheral immune cells; SHAP

Key Summary Points

Why carry out this study?

Low back pain (LBP) is a leading cause of disability worldwide, contributing to substantial economic and healthcare burdens. Current treatment outcomes are highly variable, and the molecular mechanisms influencing response to treatment remain unclear

There is an unmet need for personalized treatment strategies in LBP management to predict which patients will benefit from specific interventions

This study aimed to develop machine learning models based on gene expression signatures from peripheral immune cells to predict treatment outcomes in patients with LBP

What was learned from the study?

The study identified key genes associated with treatment outcomes and demonstrated that certain machine learning models achieved high accuracy in predicting whether patients with LBP would experience persistent or resolved pain

The results indicate that gene signature-based models hold promise for informing personalized treatment strategies in LBP management. Even if some models underperformed, the identification of key molecular markers has the potential to improve future research and clinical interventions in chronic pain management

INTRODUCTION

Low back pain (LBP) is a leading cause of disability and affects millions of individuals worldwide, representing a substantial public health burden [1]. It is estimated that over 80% of the population will experience LBP at some point in their lives, with a significant proportion developing chronic or recurrent pain [2]. The multifaceted components of the lumbar spine, including soft tissues, vertebrae, zygapophyseal and sacroiliac joints, intervertebral discs, and neurovascular structures, are susceptible to diverse stressors that can, individually or synergistically, precipitate LBP [3]. According to the Global Burden of Disease Study, LBP has consistently ranked as one of the top causes of years lived with disability (YLDs) globally, contributing to both direct healthcare costs and indirect costs related to lost productivity and reduced quality of life [4]. The prevalence of LBP increases with age, and as the global population continues to age, the economic and social impact of LBP is expected to rise further [5].

Despite its high prevalence and burden, the etiology of LBP remains complex and

multifactorial, encompassing a wide range of biological, psychological, and social factors [6, 7]. LBP can be caused by structural issues such as herniated discs, spinal stenosis, and degenerative disc disease, but in many cases, the exact underlying cause remains unclear [8]. This uncertainty in diagnosis and the variability in patient response to treatments complicate the management of LBP. Current treatment strategies for LBP include physical therapy, pharmacological interventions, and, in severe cases, surgical options [9]. However, these treatments often yield mixed results, and a significant proportion of patients continue to experience persistent pain despite intervention. The unpredictable nature of treatment outcomes for LBP underscores the need for personalized approaches to pain management. While certain patients may respond well to conventional treatments, others may experience little to no improvement, suggesting that individual biological factors play a critical role in determining treatment efficacy. Understanding the molecular mechanisms underlying LBP and identifying biomarkers that can predict treatment response are key challenges in the field.

Advances in bioinformatics and machine learning techniques offer new opportunities to address these challenges by analyzing large-scale molecular data to uncover the biological signatures associated with different pain trajectories [10, 11]. One promising avenue for research is the analysis of transcriptomic data from peripheral immune cells. Increasing evidence suggests that the immune system plays a crucial role in the pathophysiology of chronic pain, including LBP [12]. Immune cells, particularly those involved in inflammatory processes, can modulate pain pathways through the release of cytokines and other signaling molecules [13, 14]. By studying the gene expression profiles of peripheral immune cells, researchers can gain insights into the molecular mechanisms that may contribute to the persistence or resolution of pain in patients with LBP. Additionally, using machine learning models to analyze gene expression data has the potential to identify key genes associated with treatment response, paving the way for the development of predictive models that can inform personalized treatment strategies for LBP.

Given the significant global burden of LBP and the variability in treatment outcomes, this study aims to develop a gene signature-based machine learning model using transcriptomic data from peripheral immune cells to predict treatment outcomes in patients with LBP. In this study, we developed a comprehensive machine learning approach to predict treatment outcomes in patients with LBP using transcriptomic data from peripheral immune cells. The overall workflow consisted of several key steps. First, patients with LBP were classified into two groups based on their response to treatment after 3 months: those whose pain resolved and those with persistent pain. Differential gene expression analysis was performed to identify key genes associated with these treatment outcomes. Next, five machine learning models—Lasso, Elastic Net, Random Forest, Support Vector Machine (SVM), and Gradient Boosting Machine (GBM)—were employed to select core genes for model construction. These core genes were then used in combination with nine different machine learning algorithms, including Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and others, resulting in a total of 45 model combinations. The predictive performance of these models was evaluated based on accuracy, and SHapley Additive exPlanations (SHAP) analysis was used to interpret the contribution of each gene to the model's predictions. This scheme allowed us to systematically identify the key genetic factors influencing treatment outcomes in patients with LBP and provided a robust framework for building predictive models to inform personalized treatment strategies. The detailed procedure is illustrated in Fig. 1.

METHODS

Data Source

The transcriptomic data of patients with LBP from peripheral immune cells was retrieved from the GEO database (GSE177034). This

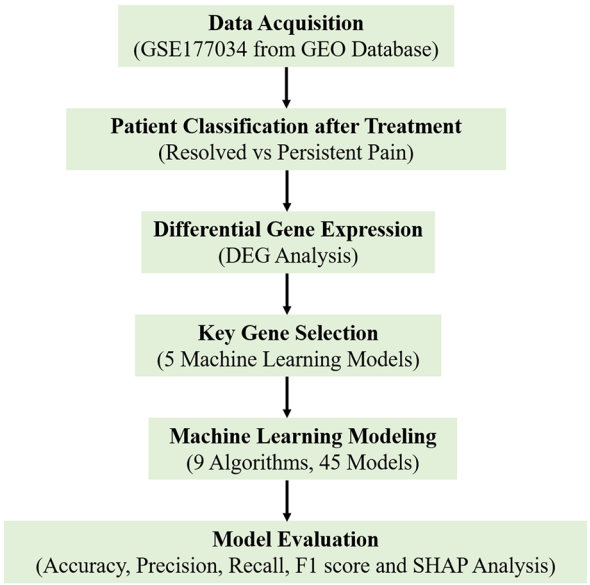


Fig. 1 Flowchart of machine learning model for predicting treatment outcomes in LBP patients based on gene signatures. *DEG* differentially expressed gene, *GEO* Gene Expression Omnibus, *SHAP*, SHapley Additive exPlanations

study utilized transcriptomic data from a human low back pain (LBP) cohort, part of the PainOMICs project (NCT02037763), funded by the European Community under FP7. The primary aim was to examine associations between genome-wide transcriptomics and the development of persistent chronic LBP. A retrospective selection of the first 50 patients with resolved pain and 50 patients with persistent pain, all white adults, was made. Inclusion criteria included acute LBP (pain level ≥ 4 on NRS) of < 6-week duration, while exclusion criteria ruled out a recent history of persistent chronic LBP, spinal fractures, and unstable disease. Patients received standardized treatment, including non-steroidal anti-inflammatory drug (NSAID), steroids, and opioids, based on clinical signs. Follow-up at 3 months classified patients into "resolved pain" (NRS < 4) or "persistent pain" (NRS ≥ 4) groups. The researcher conducting transcriptomic analysis was blinded to patient group assignment.

Ethical Approval

This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by any of the authors. The transcriptomic data of patients with LBP from peripheral immune cells were retrieved from the GEO database (GSE177034). This study utilized transcriptomic data from a human low back pain (LBP) cohort, part of the PainOMICs project (NCT02037763), funded by the European Community under FP7. All procedures performed in those studies involving human participants were approved by the University Hospital of Parma's Ethics Committee (protocol number 43543).

Differentially Expressed Gene Analysis

To identify differentially expressed genes (DEGs) between the persistent and resolved pain groups, we performed a differential expression analysis using the normalized gene expression data [15]. First, the raw transcriptomic data from the GSE177034 dataset were imported and normalized using the normalizeBetweenArrays function from the limma package to ensure comparability across samples. We then created a grouping variable (group_info) that labeled samples as either "Control" (persistent pain) or "Test" (resolved pain) based on their clinical status. A linear model was fitted to the normalized expression data using the lmFit function from the limma package, followed by empirical Bayes moderation with the eBayes function to improve the statistical power of the analysis [16]. Finally, the topTable function was used to extract the list of DEGs, applying the Benjamini-Hochberg (BH) method for multiple testing correction. Genes with an adjusted *p*-value (p.adj) < 0.1 were considered differentially expressed. The analyses were conducted in R (version 4.3.3).

Key Genes Selected

To identify key genes for model construction, we employed multiple machine learning

techniques for feature selection using the expression data. The dataset was split into training and testing sets with an 80:20 ratio [17]. The gene expression data were standardized using the StandardScaler function to ensure that features were on a comparable scale. We applied five machine learning models—Lasso, Elastic Net, Random Forest, Support Vector Machine (SVM), and Gradient Boosting Machine (GBM)—to select the most important genes. The models were trained on the standardized training data, and feature selection was performed using the SelectFromModel function [18]. For Lasso and Elastic Net, hyperparameters were set with alpha = 0.01 to ensure regularization, while Elastic Net had an l1_ratio of 0.5 to balance L1 and L2 penalties. Random Forest and GBM were implemented with 100 estimators, and SVM was used with a linear kernel to perform gene selection. Each model identified a set of genes that were important for classifying patients with LBP into "resolved pain" or "persistent pain" groups. Ultimately, these genes were designated as the core features for the subsequent predictive model construction. The final list of key genes was saved for further analysis. The analyses were conducted in Python (version 3.12).

Machine Learning Modeling

To predict treatment outcomes in patients with LBP, we applied nine different machine learning classifiers to the gene sets selected by five feature selection methods. The classifiers used were Logistic Regression, K-Nearest Neighbors, Support Vector Machine (SVM), Decision Tree, Random Forest, Gradient Boosting Machine (GBM), Multilayer Perceptron (MLP), Naive Bayes, and Linear Discriminant Analysis (LDA). This resulted in a total of 45 machine learning model combinations. For each feature selection method, the gene subsets were extracted, and the corresponding gene expression values were used as input features for model training. To ensure robust model evaluation and minimize overfitting, we implemented five-fold cross-validation. In this process, the dataset was randomly divided into five equal parts, with each part serving as the validation set once while the remaining four parts were used for training. The features were standardized using StandardScaler to ensure that each feature had a mean of 0 and a standard deviation of 1. Each classifier was then trained on the standardized training data, and the models were evaluated on the test set. The analyses were conducted in Python (version 3.12).

Machine Learning Models Evaluation

The performance of each model was measured using accuracy, precision, recall, and F1 score [19]. This comprehensive machine learning modeling approach allowed us to systematically compare the performance of different models in predicting LBP treatment outcomes based on the selected gene signatures. The SHapley Additive exPlanations (SHAP) method was used to visualize the contribution of core genes to the model of the first fold cross performance, highlighting their roles in predicting treatment outcomes [20]. The confusion matrix was used to visualize the predictive performance of the model of the first fold cross on the validation set.

RESULTS

Analysis of Differentially Expressed Genes and Screening of Key Genes

As illustrated in Fig. 2A, a total of 61 DEGs were identified between patients with resolved and persistent pain.

Following the application of five machine learning models—Lasso, Elastic Net, Random Forest, Support Vector Machine (SVM), and Gradient Boosting Machine (GBM)—a set of core genes was identified for each feature selection method. These models were applied to select key genes from the transcriptomic data of patients with resolved and persistent LBP. The results showed that certain genes, such as DNASE1L2, CCDC85B, and PTTG1, were consistently selected across multiple methods, indicating their potential significance in distinguishing between the two groups. For example,

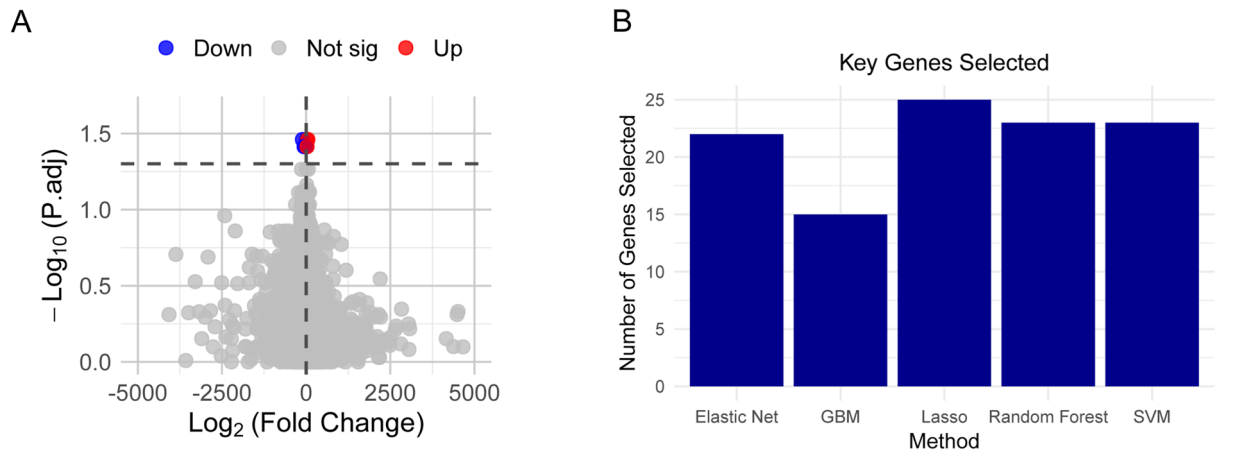


Fig. 2 Screening of differentially expressed genes and key genes. **A** Differentially expressed genes between patients with resolved and persistent pain. **B** Results of key gene selection across different models. *SVM* support vector machine, *GBM* gradient boosting machine

DNASE1L2 was identified as a key gene by all five feature selection methods, while CCDC85B was selected by four of the models (Elastic Net, SVM, GBM, and Lasso). Other genes, such as SLC7A9 and H2AX, were identified by Lasso, Elastic Net, and SVM, suggesting their possible role in influencing treatment outcomes. The detailed results of the selected genes from each model are provided in Table 1 and Fig. 2B. These core genes were subsequently used in the construction of machine learning models to predict LBP treatment outcomes, forming the basis for the predictive analysis in the next phase of the study.

Machine Learning Models Evaluation

A total of 45 machine learning model combinations were constructed by combining five feature selection methods (Lasso, Elastic Net, Random Forest, Support Vector Machine, and Gradient Boosting Machine) with nine different classifiers: Logistic Regression, K-Nearest Neighbors, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting Machine, Multilayer Perceptron, Naive Bayes, and Linear Discriminant Analysis. These models were evaluated based on four performance metrics: accuracy, precision, recall, and F1 score.

As illustrated in Fig. 3, the results of our five-fold cross-validation showed that certain model combinations performed significantly better than others. The combination of Elastic Net with Logistic Regression achieved the highest accuracy of 88.7% (±8.0%), followed closely by Elastic Net with Linear Discriminant Analysis at 88.7% (±7.5%), indicating strong and stable predictive power. The Lasso and Elastic Net feature selection methods, when combined with Multilayer Perceptron, both achieved accuracies of 87.7%, with standard deviations of ±6.7% and ±9.2% respectively.

The top-performing models demonstrated robust performance across all folds, suggesting reliable generalization capability. For instance, the Lasso with Linear Discriminant Analysis showed consistent performance with an accuracy of 87.7% (±5.1%), exhibiting the lowest standard deviation among the top models, indicating stable predictive power. Notably, Multilayer Perceptron consistently appeared among the top performers when combined with various feature selection methods (Lasso, Elastic Net, and SVM), achieving accuracies > 85%.

In contrast, some models showed less effective performance, particularly those involving Decision Trees and K-Nearest Neighbors, with accuracies ranging from 63.9 to 77.4%. The combination of SVM with Decision Tree showed the lowest accuracy at 63.9% (±13.6%), with

Table 1 Detailed content of key genes in different models

Lasso	Elastic net	Random forest	SVM	GBM
DNASE1L2	DNASE1L2	DNASE1L2	DNASE1L2	DNASE1L2
H2AX	H2AX	NDUFS6	PTTG1	CCDC85B
PTTG1	PTTG1	PHKA2	CCDC85B	SDF2L1
CCDC85B	CCDC85B	PSMC3	SLC7A9	FAM184B
SLC7A9	SLC7A9	MRPL23	SPIDR	DHRS4-AS1
SDF2L1	SDF2L1	PTTG1	FAM184B	PDCL3
MTFP1	DHRS4-AS1	SLC7A9	LRRC20	DOHH
DHRS4-AS1	CLK4	SDF2L1	DHRS4-AS1	TBRG1
CLK4	ALG13	FAM184B	PDCL3	HDDC3
ALG13	TBRG1	THAP11	ALG13	NIPBL-DT
CTU2	C17orf49	DOHH	TBRG1	EPOP
HDDC3	CTU2	TBRG1	C17orf49	MIR3164
BOLA3	HDDC3	C17orf49	HDDC3	LOC100996506
NPEPPSP1	BOLA3	LINC00266-1	BOLA3	LOC106780825
NIPBL-DT	NIPBL-DT	ZNF431	NIPBL-DT	LOC107984754
EPOP	MIR3164	HDDC3	EPOP	
MIR3164	LOC100996506	PTMAP11	MIR3164	
LOC100996506	LOC101930307	EPOP	LOC100996506	
ATP6V0D1-DT	UFD1-AS1	MIR3164	LOC101930307	
LOC101930307	LOC106780825	ATP6V0D1-DT	UFD1-AS1	
UFD1-AS1	LOC107984754	LOC106780825	LOC106780825	
LOC106780825	ARHGAP27P1-BPTFP1-KPNA2P3	LOC107984433	LOC107984754	
LOC107984433		LOC107984754	ARHGAP27P1-BPTFP1-KPNA2P3	
LOC107984754				
ARHGAP27P1-BPTFP1-KPNA2P3				

high variability across folds. This variability highlights the importance of selecting appropriate classifier and feature selection method combinations.

Overall, our cross-validation results revealed that Elastic Net-based and Lasso-based models, particularly when combined with Logistic Regression, Linear Discriminant Analysis, or Multilayer Perceptron, provided the most reliable predictions for low back pain treatment outcomes. The consistent performance across multiple folds suggests these models could be effective tools for predicting clinical responses in patients with LBP. The detailed results of model performance across all combinations are provided in Fig. 4 and Supplementary Table 1.

Accuracy	Precision	Recall	F1.Score	
0.85	0.83	0.84	0.83	Lasso+Logistic Regression
0.74	0.75	0.77	0.72	Lasso+K-Nearest Neighbors
0.82	0.83	0.78	0.79	Lasso+Support Vector Machine
0.67	0.64	0.69	0.64	Lasso+Decision Tree
0.73	0.76	0.68	0.69	Lasso+Random Forest
0.77	0.74	0.79	0.74	Lasso+Gradient Boosting Machine
0.88	0.89	0.84	0.86	Lasso+Multilayer Perceptron
0.70	0.67	0.76	0.69	Lasso+Naive Bayes
0.88	0.84	0.91	0.87	Lasso+Linear Discriminant Analysis
0.89	0.89	0.90	0.88	Elastic Net+Logistic Regression
0.76	0.74	0.85	0.76	Elastic Net+K-Nearest Neighbors
0.83	0.81	0.88	0.83	Elastic Net+Support Vector Machine
0.66	0.64	0.59	0.60	Elastic Net+Decision Tree
0.77	0.77	0.84	0.77	Elastic Net+Random Forest
0.73	0.76	0.71	0.70	Elastic Net+Gradient Boosting Machine
0.88	0.90	0.86	0.87	Elastic Net+Multilayer Perceptron
0.64	0.64	0.74	0.65	Elastic Net+Naive Bayes
0.89	0.87	0.91	0.88	Elastic Net+Linear Discriminant Analysis
0.82	0.80	0.84	0.82	Random Forest+Logistic Regression
0.75	0.75	0.81	0.74	Random Forest+K-Nearest Neighbors
0.82	0.82	0.79	0.79	Random Forest+Support Vector Machine
0.73	0.72	0.73	0.71	Random Forest+Decision Tree
0.79	0.80	0.78	0.77	Random Forest+Random Forest
0.82	0.82	0.82	0.80	Random Forest+Gradient Boosting Machine
0.83	0.80	0.84	0.81	Random Forest+Multilayer Perceptron
0.71	0.67	0.76	0.70	Random Forest+Naive Bayes
0.75	0.72	0.79	0.74	Random Forest+Linear Discriminant Analysis
0.85	0.87	0.84	0.85	SVM+Logistic Regression
0.68	0.68	0.76	0.67	SVM+K-Nearest Neighbors
0.85	0.84	0.88	0.85	SVM+Support Vector Machine
0.64	0.63	0.64	0.61	SVM+Decision Tree
0.79	0.81	0.76	0.77	SVM+Random Forest
0.80	0.77	0.83	0.78	SVM+Gradient Boosting Machine
0.86	0.83	0.86	0.84	SVM+Multilayer Perceptron
0.68	0.65	0.79	0.68	SVM+Naive Bayes
0.84	0.76	0.87	0.81	SVM+Linear Discriminant Analysis
0.83	0.80	0.80	0.79	GBM+Logistic Regression
0.73	0.75	0.77	0.71	GBM+K-Nearest Neighbors
0.77	0.81	0.73	0.75	GBM+Support Vector Machine
0.77	0.79	0.79	0.76	GBM+Decision Tree
0.80	0.82	0.79	0.78	GBM+Random Forest
0.75	0.77	0.75	0.72	GBM+Gradient Boosting Machine
0.85	0.84	0.86	0.83	GBM+Multilayer Perceptron
0.72	0.69	0.77	0.71	GBM+Naive Bayes
0.84	0.80	0.82	0.80	GBM+Linear Discriminant Analysis



◂**Fig. 3** Different model combinations evaluated. *SVM* support vector machine, *GBM* gradient boosting machine

Contribution of Core Genes to Model Performance

The SHAP method was used to visualize the contribution of core genes to model performance (as illustrated in Fig. 5). The SHAP analysis across 13 top-performing machine learning models identified key genes consistently contributing to predictions of low back pain (LBP) treatment outcomes. Genes such as DNASE1L2, CCDC85B, SDF2L1, UFD1-AS1, and H2AX were repeatedly highlighted across models, suggesting their roles in immune regulation, cellular stress, and DNA repair processes, all of which are likely critical in distinguishing between patients with resolved and persistent pain. These genes serve as potential biomarkers for predicting treatment outcomes, offering opportunities for personalized interventions in LBP management. Additionally, model-specific genes like NIPBL-DT further emphasize the molecular diversity involved in pain persistence. This comprehensive analysis provides a strong foundation for future research into tailored pain treatments. The ability to accurately predict LBP treatment outcomes using gene signatures from peripheral immune cells opens avenues for personalized medicine. Identifying patients likely to experience persistent pain could lead to early intervention and tailored treatment strategies, potentially improving clinical outcomes and reducing healthcare costs.

DISCUSSION

The Elastic Net with Logistic Regression achieved the highest accuracy of 88.7% ± 8.0% (mean ± standard deviation), followed closely by Elastic Net with Linear Discriminant Analysis (88.7% ± 7.5%) and Lasso with Multilayer Perceptron (87.7% ± 6.7%). These models effectively predicted treatment outcomes and distinguishing between patients who responded to treatment and those with persistent pain. Furthermore, SHAP analysis provided valuable insights into the contribution of individual genes to model predictions, revealing that core genes like DNASE1L2, CCDC85B, SDF2L1, UFD1-AS1, and H2AX had significant impacts on model performance. These findings underscore the potential of gene expression data and machine learning models in advancing personalized medicine for LBP, enabling earlier intervention and tailored treatment strategies.

The consistent selection of certain genes across multiple machine learning methods indicates their potential biological importance in distinguishing between patients with resolved and persistent pain. Of particular significance, DNASE1L2 was identified by all five feature selection methods, suggesting a robust role in the underlying pathophysiology of pain resolution. DNASE1L2, a member of the DNase family, is involved in apoptotic DNA degradation and plays a crucial role in the clearance of cell debris during inflammation, which is particularly relevant as efficient clearance of inflammatory mediators is essential for preventing chronic inflammation and subsequent pain persistence [21, 22]. CCDC85B and PTTG1, selected by four methods, represent another layer of molecular complexity in pain chronicity. CCDC85B has been implicated in the regulation of Wnt signaling and NF-κB pathways, both central to inflammatory responses and pain sensitization, while PTTG1, primarily known as a securin protein involved in cell cycle regulation, has recently been linked to immune cell proliferation and cytokine production [23–26]. Additionally, our analysis identified several genes involved in immune cell function and inflammation resolution, such as CD28 and ICOS for T-cell activation and differentiation, and IL1R1 and TNFRSF members for cytokine signaling pathways, emphasizing the critical role of immune-mediated inflammation in determining pain outcomes [27–29]. The selection of genes involved in cellular stress responses and tissue repair (including heat shock protein family members and growth factor-related genes) suggests that the capacity for cellular adaptation and tissue homeostasis may be key determinants in pain resolution [30–32]. Pathway analysis of the identified genes reveals enrichment

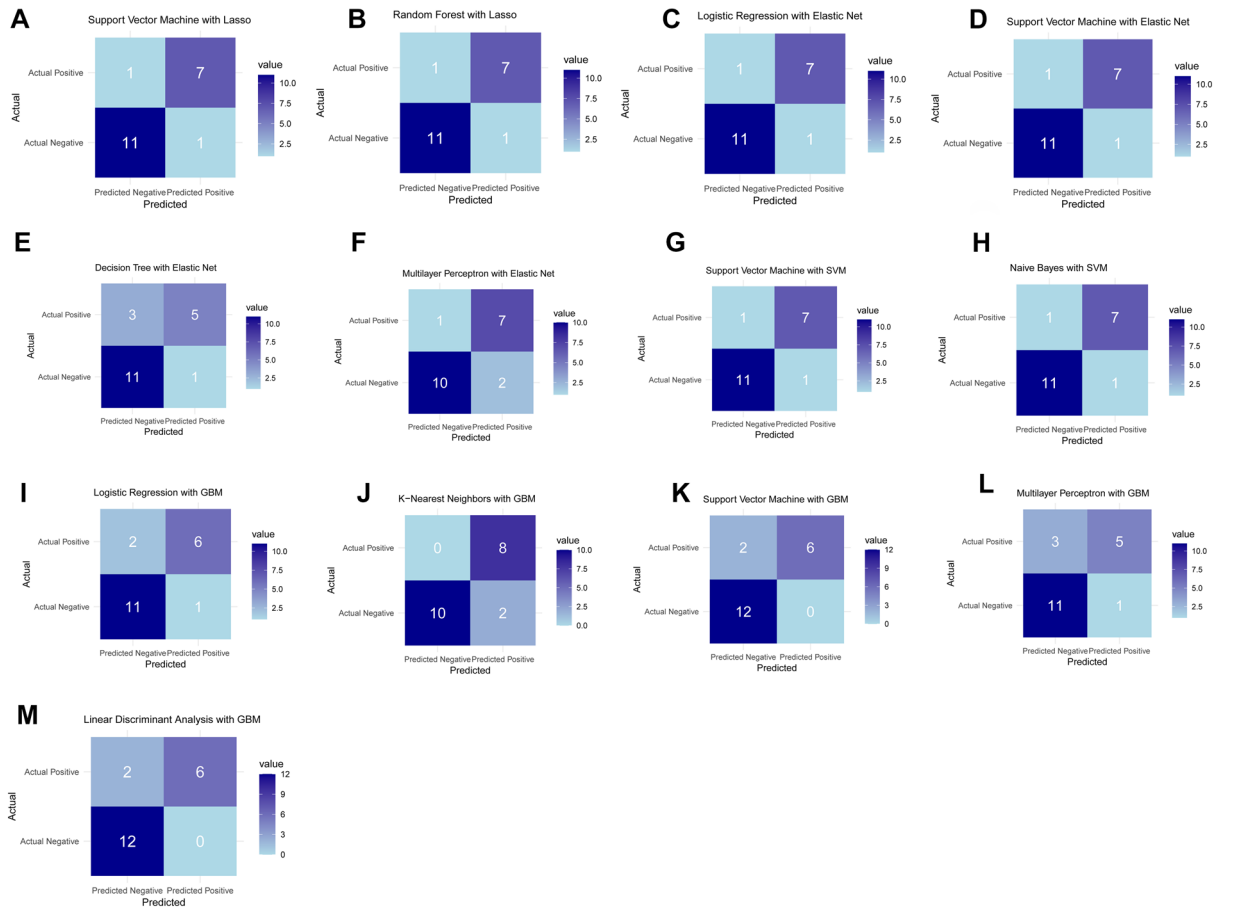


Fig. 4 Evaluation of different machine learning models. **A** Support vector machine with Lasso. **B** Randon Forest with Lasso. **C** Logistic Regression with Elastic Net. **D** Support Vector Machine with Elastic Net. **E** Decision Tree with Elastic Net. **F** Multilayer Perceptron with Elastic Net. **G** Support Vector Machine with SVM. **H** Naive Bayes with SVN. **I** Logistic Regression with GBM. **J** K-Nearest Neighbors with GBM. **K** Support Vector Machine with GBM. **L** Multilayer Perceptron with GBM. **M** Linear Discriminant Analysis with GBM. *SVM* support vector machine, *GBM* gradient boosting machine

in several key biological processes, including inflammatory response regulation, cellular stress response and adaptation, immune cell activation and differentiation, apoptotic regulation, tissue repair, and cytokine signaling modulation. These core genes likely represent key molecular signatures that could offer insights into the immune mechanisms driving chronic pain or its resolution, with their identification through multiple machine learning approaches not only validating their biological significance but also highlighting the utility of transcriptomic data in uncovering biological pathways and markers that are not immediately apparent through traditional clinical assessments. The convergence of multiple feature selection methods on these genes strengthens the evidence for their importance in pain chronicity and suggests potential therapeutic targets for intervention in chronic pain conditions.

Furthermore, the study demonstrated that certain machine learning model combinations performed exceptionally well, particularly those involving Elastic Net and Lasso combined with Logistic Regression, Linear Discriminant Analysis, and Multilayer Perceptron. Through five-fold cross-validation, these models consistently achieved accuracies > 87%, with relatively low

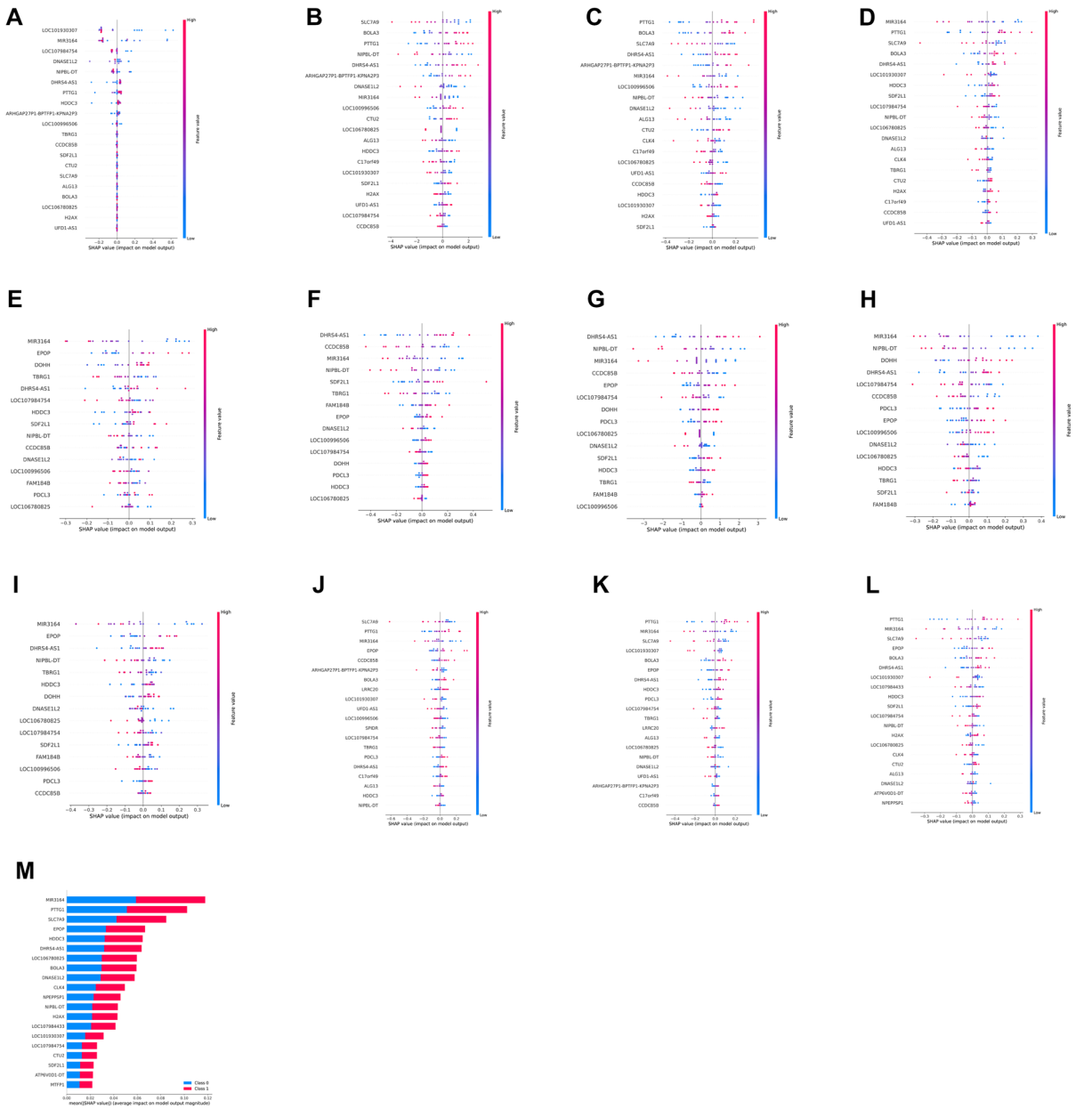


Fig. 5 The SHAP analysis of machine learning models for core gene selection. **A** Decision Tree with Elastic Net. **B** Logistic Regression with Elastic Net. **C** Multilayer Perceptron with Elastic Net. **D** Support Vector Machine with Elastic Net. **E** Nearest Neighbors with GBM. **F** Linear Discriminant Analysis with GBM. **G** Logistic Regression with GBM. **H** Multilayer Perceptron with GBM. **I** Support Vector Machine with GBM. **J** Naive Bayes with SVM. **K** Support Vector Machine with SVM. **L** Support Vector Machine with Lasson. **M** Random Forest with Lasso. *SVM* support vector machine, *GBM* gradient boosting machine, *SHAP* SHapley Additive exPlanations

standard deviations, indicating robust and reliable performance in predicting treatment outcomes based on gene expression data. The ability to accurately classify patients into those who will experience persistent pain versus those who will achieve pain resolution is of great clinical significance, as it could help clinicians tailor interventions more effectively. The success of

the Elastic Net and Logistic Regression combination (88.7% ± 8.0%) can be attributed to Elastic Net's ability to perform variable selection while handling correlated features, effectively identifying the most relevant genes, and Logistic Regression's capacity to provide stable, interpretable results with good generalization [33, 34]. Similarly, the strong performance of Elastic Net with Linear Discriminant Analysis (88.7% ± 7.5%) suggests that these methods are particularly effective at capturing the underlying patterns in gene expression data that predict clinical outcomes in LBP. The Multilayer Perceptron showed consistent high performance when combined with different feature selection methods, including Lasso (87.7% ± 6.7%), Elastic Net (87.7% ± 9.2%), and SVM (85.7% ± 8.5%), demonstrating its robust capability in handling complex gene expression patterns. This consistency across different feature selection methods suggests that neural network-based approaches might be particularly suited for capturing subtle relationships in transcriptomic data. On the other hand, models involving Decision Trees and K-Nearest Neighbors consistently underperformed, with accuracies ranging from 63.9 to 77.4%. Decision Tree-based models showed particularly high variability in performance, with standard deviations > 13% in some cases, highlighting their instability with high-dimensional gene expression data. This notable difference in performance underscores the critical importance of selecting appropriate machine learning algorithms when working with transcriptomic data for clinical prediction tasks.

The high accuracy and robustness of certain models, combined with the biological relevance of the identified genes, suggest that these gene signature-based models could have significant clinical applications. In practice, such models could be used as part of a diagnostic tool to predict whether a patient is likely to respond to a given treatment or whether their pain is likely to persist, allowing for more personalized treatment strategies. For example, patients predicted to have persistent pain could be offered more aggressive or targeted interventions early in their treatment course, potentially preventing chronic pain development. In contrast, those predicted to have resolved pain might benefit from a more conservative approach, minimizing unnecessary treatments and reducing healthcare costs. The integration of gene expression data with machine learning models represents a significant step towards personalized medicine in the context of LBP. By identifying specific molecular signatures associated with treatment outcomes, clinicians could move away from the current "one-size-fits-all" approach to a more tailored method of pain management. This would enable the design of individualized treatment plans that consider each patient's unique molecular profile, leading to improved outcomes and a reduction in the burden of chronic pain.

Limitations

Several limitations must be acknowledged. First, while the use of gene expression data from peripheral immune cells offers valuable insights into the molecular mechanisms of low back pain (LBP), this approach is limited by the inherent complexity of LBP as a multifactorial condition. The study focused primarily on immune-related gene signatures, potentially overlooking other biological processes and tissue-specific pathways, such as those related to the musculoskeletal system, that could also play significant roles in pain persistence and resolution. Second, the sample size, although adequate for initial exploration, may limit the generalizability of the findings. The relatively small number of patients in each group (resolved vs. persistent pain) may have reduced the statistical power to detect subtle, yet clinically important, gene-expression differences. Larger, independent cohorts are needed to validate the identified gene signatures and the predictive accuracy of the machine learning models. Third, the study exclusively included white patients, which limits the generalizability of the findings to other ethnic groups. Genetic diversity across populations can influence both gene expression patterns and the effectiveness of predictive models. Future studies should include more diverse populations to assess whether the identified gene signatures are universally applicable or population-specific. Fourth, while multiple machine learning models were employed to predict treatment outcomes, there is a risk

of overfitting, particularly with more complex models such as Random Forest and Gradient Boosting Machine. Overfitting can lead to models that perform well on training data but fail to generalize to new, unseen data. Although we applied cross-validation to mitigate this, further validation on independent datasets is necessary. Lastly, the study focused on predicting outcomes after 3 months of treatment. While this timeframe is relevant for assessing short- to mid-term responses, it does not capture long-term outcomes or the potential for pain recurrence, which is common in LBP. Longitudinal studies are needed to evaluate whether these gene signatures can predict sustained treatment success over longer periods.

CONCLUSION

This study provides a robust framework for using transcriptomic data and machine learning models to predict treatment outcomes in patients with LBP. The identification of key genes and the development of highly accurate predictive models highlight the potential for these approaches to be integrated into clinical practice. As further validation is conducted and models are refined, this work paves the way for more personalized and effective treatments for LBP, ultimately improving patient care and outcomes.

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Data Availability. The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request. The transcriptomic data of patients with LBP from peripheral immune cells was retrieved from the GEO database (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE177034). We fully support open science and transparency in research. As suggested, we have made our complete codebase publicly available in the Supplementary Materials. This includes all scripts used for data processing and analysis, allowing for full reproducibility of our results.

Declarations

Conflict of Interest. The authors (Youzhi Lian, Yinyu Shi, Haibin Shang, and Hongsheng Zhan) declare no conflict of interest.

Ethical Approval. This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by any of the authors. The transcriptomic data of patients with LBP from peripheral immune cells was retrieved from the GEO database (GSE177034). This study utilized transcriptomic data from a human low back pain (LBP) cohort, part of the PainOMICs project (NCT02037763), funded by the European Community under FP7. All procedures performed in those studies involving human participants were approved by the University Hospital of Parma's Ethical Committee (Protocol Number 43543).

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REFERENCES

1. Negrini S, Pollet J, Ranica G, Donzelli S, Vanossi M, Piovanelli B, et al. Movement analysis could help in the assessment of chronic low back pain patients: results from a preliminary explorative study. Int J Environ Res Public Health. 2022;19:9033.

2. Lin C-Y, Liu Y-H, Chen S-M, Cheng S-C, Liu M-F. The effectiveness of group-based core stability exercise and educational booklet for hospital workers in Taiwan with nonspecific low back pain: a preliminary study. Int J Environ Res Public Health. 2022;19:3324.

3. Knezevic NN, Candido KD, Vlaeyen JWS, Zundert JV, Cohen SP. Low back pain. The Lancet. 2021;398:78–92.

4. Gorasso V, Van der Heyden J, De Pauw R, Pelgrims I, De Clercq EM, De Ridder K, et al. The health and economic burden of musculoskeletal disorders in Belgium from 2013 to 2018. Popul Health Metr. 2023;21:4.

5. Bryndal A, Glowinski S, Majchrzycki M. Influence of pregnancy on the occurrence of lumbar spine pain in polish women: a retrospective study. J Pers Med. 2022;12:357.

6. Liu H, Li H, Huang L, Tian H, Wu J, Guan Q, et al. The link between childhood physical activity enjoyment and adult kinesiophobia in individuals with chronic low back pain. BMC Public Health. 2024;24:2557.

7. Zhang T, Firouzabadi A, Yang D, Liu S, Schmidt H. Age-dependent flexion relaxation phenomenon in chronic low back pain patients. Front Bioeng Biotechnol. 2024;12:1388229.

8. Chou R. Low back pain. Ann Intern Med. 2021;174:ITC113–28.

9. Vlaeyen JWS, Maher CG, Wiech K, Van Zundert J, Meloto CB, Diatchenko L, et al. Low back pain. Nat Rev Dis Primers. 2018;4:52.

10. Rashid J, Batool S, Kim J, Wasif Nisar M, Hussain A, Juneja S, et al. An augmented artificial intelligence approach for chronic diseases prediction. Front Public Health. 2022;10: 860396.

11. Zhang X, Zhang H, Fan C, Hildesjö C, Shen B, Sun X-F. Loss of CHGA protein as a potential biomarker for colon cancer diagnosis: a study on biomarker discovery by machine learning and confirmation by immunohistochemistry in colorectal cancer tissue microarrays. Cancers (Basel). 2022;14:2664.

12. Singh SK, Krukowski K, Laumet GO, Weis D, Alexander JF, Heijnen CJ, et al. CD8+ T cell–derived IL-13 increases macrophage IL-10 to resolve neuropathic pain. JCI Insight. 2022;7:e154194.

13. Bermudez-Lekerika P, Crump KB, Tseranidou S, Nüesch A, Kanelis E, Alminnawi A, et al. Immunomodulatory effects of intervertebral disc cells. Front Cell Dev Biol. 2022;10: 924692.

14. Artero NA, Manchope MF, Carvalho TT, Saraiva-Santos T, Bertozzi MM, Carneiro JA, et al. Hesperidin methyl chalcone reduces the arthritis caused by TiO_2 in mice: targeting inflammation, oxidative stress, cytokine production, and nociceptor sensory neuron activation. Molecules. 2023;28:872.

15. Bresciani E, Squillace N, Orsini V, Piolini R, Rizzi L, Molteni L, et al. miRNA expression profiling in subcutaneous adipose tissue of monozygotic twins discordant for HIV infection: validation of differentially expressed miRNA and bioinformatic analysis. Int J Mol Sci. 2022;23:3486.

16. Alvarez M, Benhammou JN, Darci-Maher N, French SW, Han SB, Sinsheimer JS, et al. Human liver single nucleus and single cell RNA sequencing identify a hepatocellular carcinoma-associated cell-type affecting survival. Genome Med. 2022;14:50.

17. Poongavanam V, Kölling F, Giese A, Göller AH, Lehmann L, Meibom D, et al. Predictive modeling of PROTAC cell permeability with machine learning. ACS Omega. 2023;8:5901–16.

18. Shaik S, Singh A, Suresh A, Ahmed N. Genome Informatics and machine learning-based identification of antimicrobial resistance-encoding features and virulence attributes in *Escherichia coli* genomes representing globally prevalent lineages, including high-risk clonal complexes. MBio. 2022;13: e03796-21.

19. Ahmad B, Sun J, You Q, Palade V, Mao Z. Brain tumor classification using a combination of

variational autoencoders and generative adversarial networks. Biomedicines. 2022;10:223.

20. Divate M, Tyagi A, Richard DJ, Prasad PA, Gowda H, Nagaraj SH. Deep learning-based pan-cancer classification model reveals tissue-of-origin specific gene expression signatures. Cancers (Basel). 2022;14:1185.

21. Delfino D, Mori G, Rivetti C, Grigoletto A, Bizzotto G, Cavozzi C, et al. Actin-resistant DNase1L2 as a potential therapeutics for CF lung disease. Biomolecules. 2021;11:410.

22. Wu Z, Liu Q. Effects of repetitive transcranial magnetic stimulation combined with acupuncture on NLRP3 inflammasome and protease levels in patients with neuropathic pain. Am J Transl Res. 2023;15:4699–708.

23. Feng Y, Gao Y, Yu J, Jiang G, Zhang X, Lin X, et al. CCDC85B promotes non-small cell lung cancer cell proliferation and invasion. Mol Carcinog. 2019;58:126–34.

24. Colloca L, Mocci E, Wang Y, Massalee R, Chen S, White J, et al. Transcriptomic profiles associated with experimental placebo effects in chronic pain. Clin Pharmacol Ther. 2024;116:380–9.

25. Park S, Kim S, Kim M-Y, Lee SS, Choi J. Pituitary tumor-transforming gene 1 regulates the senescence and apoptosis of oral squamous cell carcinoma in a p21-dependent DNA damage response manner. Oncol Rep. 2024;52:135.

26. Wu H, Li D, Zhang C-Y, Huang L-L, Zeng Y-J, Chen T-G, et al. Restoration of ARA metabolic disorders in vascular smooth muscle cells alleviates intimal hyperplasia. Eur J Pharmacol. 2024;983: 176824.

27. Sankaranarayanan I, Tavares-Ferreira D, Mwirigi JM, Mejia GL, Burton MD, Price TJ. Inducible co-stimulatory molecule (ICOS) alleviates paclitaxel-induced neuropathic pain via an IL-10-mediated mechanism in female mice. J Neuroinflammation. 2023;20:32.

28. Orvain C, Cauvet A, Prudent A, Guignabert C, Thuillet R, Ottaviani M, et al. Acazicolcept (ALPN-101), a dual ICOS/CD28 antagonist, demonstrates efficacy in systemic sclerosis preclinical mouse models. Arthritis Res Ther. 2022;24:13.

29. Kang C, Yun F, Shi L, Jia Y, Liu X. Landscape of costimulatory molecule signature in breast cancer and its prognostic significance. Ann Transl Med. 2023;11:59.

30. Besednova NN, Andryukov BG, Zaporozhets TS, Kuznetsova TA, Kryzhanovsky SP, Ermakova SP, et al. Molecular targets of brown algae phlorotannins for the therapy of inflammatory processes of various origins. Mar Drugs. 2022;20:243.

31. Żywień U, Barczyk-Pawelec K, Sipko T. Associated risk factors with low back pain in white-collar workers—a cross-sectional study. J Clin Med. 2022;11:1275.

32. Abshirini M, Coad J, Wolber FM, von Hurst P, Miller MR, Tian HS, et al. Effects of Greenshell™ mussel intervention on biomarkers of cartilage metabolism, inflammatory markers and joint symptoms in overweight/obese postmenopausal women: a randomized, double-blind, and placebo-controlled trial. Front Med (Lausanne). 2022;9:1063336.

33. Zhu-Zhou F, Gil-Pita R, García-Gómez J, Rosa-Zurera M. Robust multi-scenario speech-based emotion recognition system. Sensors (Basel). 2022;22:2343.

34. Zhu W, Lévy-Leduc C, Ternès N. Identification of prognostic and predictive biomarkers in high-dimensional data with PPLasso. BMC Bioinformatics. 2023;24:25.